Journal of Membrane Science 737 (2026) 124781

Contents lists available at ScienceDirect

Journal of Membrane Science

journal homepage: www.elsevier.com/locate/memsci

The effects of membrane structure and filtration conditions on virus filtration

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ARTICLE INFO

Keywords:
Hollow fiber
Laser scanning confocal microscopy
Minute virus of mice
Monoclonal antibody
Polymeric membrane
Irreversible aggregates
Virus clearance

ABSTRACT

Virus filtration is used for validation of virus clearance in the manufacture of monoclonal antibodies. Requirements for this size exclusion process include 10,000-fold reduction of virus particles and at least 95 % recovery of the monoclonal antibody in the permeate. The size difference between the rejected parvovirus particles, 25 nm, and monoclonal antibody, 10–12 nm, is about two. Membranes are designed with pore sizes around the average size of the parvovirus. Given the polydispersity in membrane pore size and virus particles, filter performance is dependent on membrane properties, feed and operating conditions. Compromised performance usually occurs due to product related foulants such as irreversible and reversible aggregates. Three commercial asymmetric hollow fiber membranes were investigated. The location of minute virus of mice entrapment within the membrane was determined by laser scanning confocal microscopy in the presence and absence of a monoclonal antibody or bovine serum albumin that forms irreversible aggregates. Experiments were conducted under constant flux for 12 h and constant pressure for 12.5–40 h. Membrane throughput was 900 L m^{-2} and effective parvovirus removal (LRVs above 4) was verified under all conditions tested. Lower permeate fluxes and longer filtration times led to virus capture deeper within the membrane. Moreover, product-membrane interaction at low fouling conditions led to the migration of virus particle deeper within the membrane. Product aggregates, similar in size to virus particles led to displacement of virus particles deeper within the membrane and broadening of the entrapment zone. These results provide unique insights into virus filter performance.

1. Introduction

Human therapeutics are divided into pharmaceuticals and biopharmaceuticals. Biopharmaceuticals are produced biologically rather than by chemical synthesis. A variety of therapeutics may be produced using a range of biological cells [1]. Here the focus is on mammalian cell culture which is routinely used to produce protein-based therapeutics such as monoclonal antibodies (mAbs). In the production of human therapeutics, it is essential that the product be of very high purity in order to minimize unwanted adverse side effects due to the presence of impurities. Regulatory authorities specify strict guidelines on product purity. In the case of biopharmaceutical products validation of clearance of impurities and contaminants of biological origin such as host cell proteins, host cell DNA and virus particles is essential. Validation of virus clearance is frequently a major challenge.

Biotherapeutics can be contaminated by both endogenous and adventitious viruses [2,3]. The level of virus clearance required depends on the manufacturing process. Typically, model retroviruses are used to validate endogenous virus clearance while model parvoviruses are used to validate adventitious virus clearance. Minute virus of mice (MVM) is frequently used to validate adventitious virus clearance. MVM is a parvovirus. It is one of the smallest spherical viruses. It is non-enveloped with the capsid about 25 nm in diameter and has an icosahedral symmetry [4]. Validating the clearance of MVM is challenging given the

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https://doi.org/10.1016/j.memsci.2025.124781
Received 10 September 2025; Received in revised form 1 October 2025; Accepted 2 October 2025
Available online 3 October 2025

small size and stability of the virus particles. Thus, it serves as a worst-case model for validating adventitious virus clearance [5,6]. Regulatory agencies, such as the US Food and Drug Administration, the European Union European Medicines Agency, the Australian Therapeutic Goods Administration etc. have established strict guidelines for MVM clearance [7,8]. Here the focus is on virus filtration for validation of MVM clearance. Detailed guidelines are outlined in various regulatory authority guidelines such as the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). The ICH brings together regulatory authorities and pharmaceutical industry in order to develop harmonized safety guidelines [7]. The guidelines explicitly state that in order to be considered an effective step, virus removal efficiency by virus filters must be Log Reduction Value (LRV) 4 or higher.

Virus filtration, commonly run in normal flow mode, involves a size exclusion process where virus particles are captured by the membrane, but the product of interest passes through the membrane pores. Virus filtration is appealing as a unit operation for validating virus clearance, as the virus filtration membrane could be an absolute barrier to virus particles. Today virus filtration membranes are often asymmetric polymeric membranes. The thin barrier layer faces the permeate side while the support structure act as an inline prefilter [9]. Given the hydrodynamic diameter of mAbs is around 10–12 nm they are expected to pass through the membrane pores [10,11].

Frequently, membrane-based size exclusion processes depend on an order of magnitude difference in the size of the species to be rejected in the retentate and the species to be recovered in the permeate. In the case of virus filtration this difference in size is two-fold or less making efficient separation challenging [9]. Further virus filtration membranes must be designed to obtain at least a 10,000-fold decrease in virus concentration in the permeate relative to the feed (4 LRV). For the process to be economically viable, more than 95 % product recovery is often required. Design requirements on throughput (minimum product recovered per membrane surface area) and permeate flux put further constraints on the virus filters [12]. As virus filtration occurs towards the end of the purification train, fouling of the membrane is frequently due to product related foulants such as dimers, trimers, reversible aggregates, misfolded proteins etc. [13].

The performance of virus filters is influenced by three main factors: (1) membrane properties such as morphology, porosity, pore structure, membrane material; (2) operating conditions including pressure and flux; and (3) feed properties such as buffer type, buffer conductivity and pH [14]. Membrane fouling represents a significant challenge to virus filtration performance. Macromolecules like proteins can form aggregates, self-associate, or interact with the membrane materials during downstream processing [15–17]. These interactions with the membrane surface could decrease flux, increase membrane resistance, and reduce overall process efficiency. The effect of membrane fouling on virus retention is complex and often difficult to predict.

Generally, in size exclusion-based filtration processes, the large difference in size between the species to be separated means that membrane pore sizes are far less than the size of the species to be rejected allowing for efficient separation. Membrane fouling generally modifies membrane rejection profiles leading to greater rejection of the species to be recovered in the permeate due to the effects such as cake formation, pore blocking etc. [18]. Besides product properties, earlier work [19] has shown that fouling of virus filters demonstrates a strong dependence on solution conditions. The initial fast flux decay during virus filtration can be described largely by a pore blocking mechanism and the subsequent slow decline by a cake filtration mechanism. In addition, previous work by Fan et al. [20] has shown that virus clearance is strongly affected by flux or applied pressure. At very low flux and low fouling conditions, virus breakthrough occurred for 3 out of the 4 commercial filters investigated. Earlier work [21] on virus filtration at low pressure also suggest that more virus breakthrough is likely at reduced pressure.

Suh et al. [22] noted that for constant pressure filtration, a high concentration of protein solution (the product to be recovered in the permeate) can lead to reduced virus rejection. However, this effect depends on the variation of membrane pore size through the membrane structure. Afzal and Zydney [23] confirmed this observation. Their membranes were much more symmetric than other commercially available virus filtration membranes. They ascribe this observation to internal polarization. They developed a model that accounts for both irreversible entrapment of viruses within the pores of the membrane as well as accumulation of viruses in a 'reservoir zone' of the membrane. Their work suggests that the protein (in their case hIgG) tends to compete with the virus particles for irreversible entrapment sites. The fact that the species to be recovered in the permeate is often only about 2 times smaller than the virus particles to be rejected means the detailed asymmetric structure of the virus filtration membrane plays a key role in determining the level of virus clearance during operation.

In order to obtain a better understanding of the effect of a product species on the level of virus clearance, recent efforts have focused on visualizing the location of virus entrapment. Fallahianbijan et al. [24] analyzed the capture of non-biological particles using laser scanning confocal microscopy (LSCM). They found that the distribution of the captured particles, even of the same size, was different for parvovirus rejection filters from different manufacturers indicating the importance of the detailed membrane structure. Studies by Ayano et al. [25] indicated that the location of virus-like particle capture shifts with filtration time, moving towards the permeate side. Leisi et al. [26] conducted visualization studies on MVM capture using LSCM. Differences in the location of entrapment of MVM for parvovirus filters from different manufacturers was observed. Based on these and other studies Shirataki and Wickramasinghe [27] and Shirataki et al. [28] have attempted to model virus filtration performance based on membrane structure.

The purpose of this study was to determine the effect of the presence of product species and operating conditions on the location of virus entrapment in the virus filtration membrane in order to better understand virus particle migration inside the membranes under different conditions. Three different hollow fiber virus filtration membranes have been investigated. Three different feed streams have been investigated containing fluorescently labelled MVM in a buffer, fluorescently labelled MVM in a mAb solution that is known not to form irreversible aggregates at the buffer condition, and in a BSA solution that is known to form irreversible aggregates. Experiments were conducted under both constant pressure and constant flux conditions. Further in manufacturing processes, it is common to flush the module with buffer to recover residual product. These operations can lead to virus breakthrough [29]. Consequently, after virus filtration a buffer flush was included. Through LSCM, detailed visualization studies have been conducted which provide insights into the location of virus entrapment. These results help deepen our understanding of the performance of virus filtration membranes.

2. Experimental section

2.1. Materials and chemicals

Model proteins used in this study were a mAb provided by Asahi Kasei Life Science Co (Tokyo, Japan) and bovine serum albumin (BSA) purchased from Lee BioSolutions (Maryland Heights, MO, USA). The following chemicals were used to prepare the buffer solution: potassium chloride (biotechnology grade) was purchased from VWR Life Science (Avantor, Radnor, PA, USA), tris base (molecular biology grade), ethylenediaminetetraacetic acid (EDTA, purified grade, ≥98.5 %), sodium phosphate dibasic (ReagentPlus, ≥99.0 %), sodium chloride and potassium dihydrogen phosphate (biotechnology grade) were purchased from MilliporeSigma (Burlington, MA, USA). The MVM virus stock and A9 mouse fibroblast cells (ATCC® CCL-1.4™), the corresponding host mammalian cell line, were purchased from the American Type Culture Collection (Manassas, VA, USA). Glucose-rich DMEM growth medium and advanced DMEM growth medium were purchased from Sigma-Aldrich (Burlington, MA, USA). Fetal bovine serum (FBS), and Trypan Blue Solution, 0.4 % were purchased from Gibco™, Thermo Fisher (Waltham, MA, USA). DyLight™ 633 NHS Ester Fluorescent dye, and paraformaldehyde, 4 % in PBS (4 % PFA), used for virus labeling and imaging sample preparation, were purchased from Thermo Fisher (Waltham, MA, USA). Tissue-Tek O.C.T. Compound, Sakura Finetek (Torrance, CA, USA) was used for cryostat sectioning of hollow fibers in preparation for LSCM imaging. Deionized (DI) water was produced using a Thermo Scientific Barnstead Smart2Pure UV/UF (Waltham, MA, USA).

Virus filtration experiments were conducted using three commercial virus filters: Planova™ 20N, Planova™ S20N and Planova™ BioEX provided by Asahi Kasei Life Science Co (Tokyo, Japan). Table 1 gives properties of the virus filters as well as the feed volume needed for a throughput of 900 L m^{-2}.

2.2. Virus filtration feed preparation

The mAb was received frozen at a concentration of 9.72 g L^{-1} in PBS buffer at pH 7.36. The conductivity of the solution was 15.6 mS cm^{-1}. The pH and conductivity were measured using Orion Star A214 pH/ISE benchtop meter with stand (Thermo Scientific). Before each virus filtration experiment, the mAb solutions were diluted with the same PBS buffer as the stock solution (pH 7.36, conductivity 15.6 mS cm^{-1}) to achieve the desired concentration of 5 g L^{-1}. The buffer was prepared by adding 137 mmol L^{-1} NaCl, 8.1 mmol L^{-1} Na_2HPO_4, 2.7 mmol L^{-1} KCl, and 1.5 mmol L^{-1} KH_2PO_4 into DI water. The pH of the buffer was adjusted to 7.36 using 2 mol L^{-1} phosphoric acid. BSA solutions at 5 g L^{-1} were prepared using the same PBS buffer. All feed streams mAb or BSA were prefiltered using a 0.1 μm sterile PES bottle top filter. All DI water and PBS buffer solutions were filtered using 0.22 μm PES bottle top filter before conducting virus filtration experiments. The mAb and BSA concentrations were determined by measuring the absorbance at 280 nm using a spectrophotometer (Genesys 100UV scanning system, Thermo Fisher).

2.3. MVM production and purification

The procedure used here is described by Asher et al. [30] and modified by Fan et al. [20]. Briefly, A9 cells were grown in T175 CELLSTAR® cell culture flasks (175 cm^2) (Thermo Fisher) with a complete glucose-rich DMEM growth medium supplemented with 10 % FBS. When the cells reached over 90 % confluence, the cells were inoculated with purified MVM stock at a target multiplicity of infection (MOI) of approximately 2.5 and incubated at 37 °C with 5 % CO_2 and 40 mL complete advanced DMEM production medium containing 1 % FBS. After 18–21 days from inoculation, centrifugation was conducted at 2, 000×g, 5,000 rpm for 15 min. The supernatant containing MVM was passed through a 0.22 μm bottle top filter (Thermo Fisher). Next the solution was filtered and buffer exchanged into TNE buffer (comprising 10 mM Tris, 150 mM NaCl, and 1 mM EDTA) using a 100 kDa Amicon Ultra-15 Centrifugal Filter (MilliporeSigma). The MVM was recovered in the retentate. Finally, the MVM containing retentate was ultracentrifuged (Beckman Coulter Ultracentrifuge, Brea, CA, USA). The MVM pellets were resuspended in 2 mL of TNE buffer. The MVM stock was sterile filtered using a 0.2 μm syringe filter (VWR) and stored at −80 °C until needed. The MVM stocks typically exhibited a titer ranging from 10.5 to 11.5 logCopies mL^{-1}, as determined by quantitative polymerase chain reaction (qPCR) assay. MVM spikes for virus filtration experiments were prepared by spiking approximately 300 μL or less of the stock solution into 300–1000 mL of feed solution to reach a feed titer of about 7.5 logCopies mL^{-1}.

Table 1
Details of virus filters and feed volumes used.

Virus filter	Filter Area (m^2)	Maximum Operating Pressure (MPa)	Membrane Material	Feed volume (mL)
Planova™ 20 N	0.001	0.098	Regenerated Cellulose	900
Planova™ S20 N	0.001	0.216	Regenerated Cellulose	900
Planova™ BioEX	0.00031	0.343	Hydrophilized polyvinylidene fluoride	270

2.4. Virus filtration

Prior to filtration visual leak tests (VLT) and equilibration was conducted following the manufacturer's protocol [31]. Briefly, all the virus filters were flushed with DI water and feed buffer to remove preservatives and extractables and trapped air. Figs. 1 and 2 give the experimental set up for constant pressure and constant flux filtration, respectively. For constant pressure filtration, a pressurized vessel connected to a nitrogen cylinder was used. The pressure was set to be 0.1 MPa. For constant flux filtration, a MasterFlex L/S digital pump (Cole-Parmer, VWR) was used. The flux was set at 75 L m^{-2} h^{-1}.

Three sets of experiments were run at constant pressure and flux: PBS buffer spiked with about 7.5 logCopies mL^{-1} MVM (qPCR), PBS containing either 5 g L^{-1} mAb or BSA spiked with about 7.5 logCopies mL^{-1} MVM (qPCR). The feed steams were then filtered through 0.22 μm bottle top filters (Thermo Fisher). Virus filtration was conducted with the three filters listed in Table 1. The target throughput was 900 L m^{-2}. For each run, once the target volumetric throughput was reached, the pressure was released, and the feed solution was switched to buffer solution (PBS feed buffer) within a minute to prevent the effects of process pause. The volume of buffer used is based on 10 L m^{-2} throughput. The only exception was for feed streams containing BSA during BioEX filtration. In this case, when the flux dropped below 20 L m^{-2} h^{-1} the feed was switched to the buffer solution. The buffer solution was run at the same pressure, 0.1 MPa, for constant pressure and the same flux, 75 L m^{-2} h^{-1}, for constant flux filtration as that of the feed solution. The filtrate was collected in 300 mL fractions and a buffer chase fraction. A Mettler Toledo scale (Columbus, OH, USA) connected to Balance Link software (Mettler Toledo) was used to monitor the filtration process. Data were recorded every minute and averaged every 10 min.

2.5. Size exclusion chromatography and dynamic light scattering

BSA and mAb feed concentrations were analyzed using an Agilent Infinity 1260 Series HPLC system (Agilent, Santa Clara, CA, USA) equipped with a size exclusion chromatography (SEC) column (TSK gel G3000 SWxl, 300 mm × 7.8 mm, Sigma- Aldrich, Burlington, MA, USA). The mobile phase was PBS buffer at pH 7.36 at a flow rate of 1.0 mL min^{-1}. The mAb and BSA were detected by UV absorbance at 220 nm using the HPLC system. This wavelength is frequently used because the peptide bond absorbs strongly in the far-UV range. The wavelength is advantageous for detecting low-concentration proteins and aggregates. Data acquisition and analysis were carried out using Chemstation B.02.01 software (Agilent). The percentages of monomers, dimers and trimers were determined by peak areas.

Dynamic light scattering (DLS) was used to determine the hydrodynamic diameter of the mAb and BSA at 5 g L^{-1}. The samples were initially filtered through a 0.2 μm polyethersulfone (PES) syringe filter (VWR), and 1 mL aliquots were transferred into disposable polystyrene cuvettes with a 1 cm path length (BrandTech, Essex, CT, USA). Measurements were conducted using a Delsa™ Nano particle size analyzer (Beckman Coulter, Brea, CA, USA). For each sample, 150 acquisitions were collected per run, and the experiment was repeated three times with independent aliquots. The resulting data were processed using the instrument software, providing mean values of hydrodynamic diameter

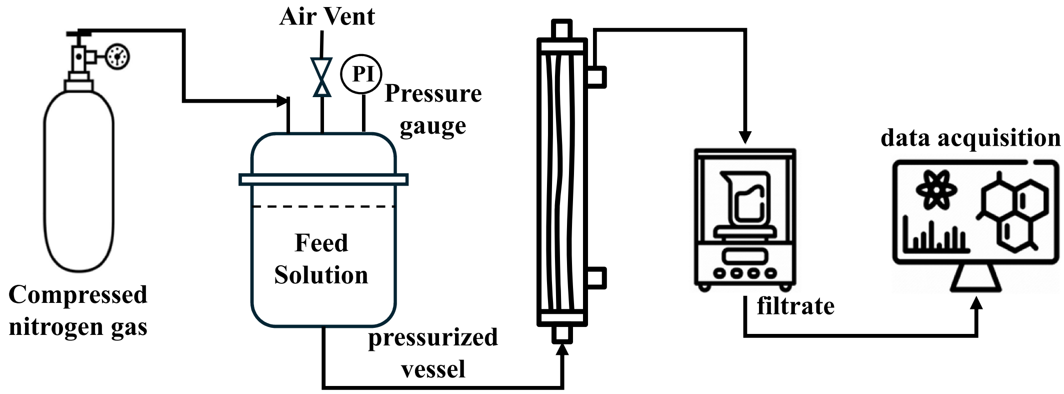

Fig. 1. Schematic diagram for constant pressure filtration at 0.1 MPa.

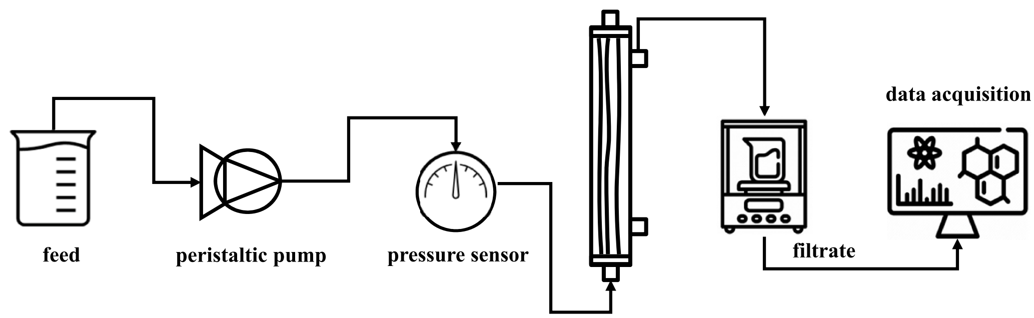

Fig. 2. Schematic diagram for constant flux filtration at 75 L m^{-2} h^{-1}.

for the mAb and BSA.

2.6. MVM titer determination

Two methods were used to determine MVM titers, qPCR and infectivity assays. The qPCR assay was used for estimating the virus titer that was spiked into the feed solution in order to prepare a feed solution with a virus titer of around 7.5 logCopies mL^{-1}. The qPCR assay measures both infective and noninfective viral particles. The infectivity assay only measures infective viral particles. The infectivity assay was used to determine LRV. The procedure described by Chen et al. [32] was used for the qPCR assay.

The $TCID_{50}$ assay (50 % tissue culture infective dose per mL) was used in this study. The indicator cell line NB324K, was donated by Peter Tattersall (Yale University). Detailed information on culturing these cells is provided by Fan et al. [20]. Trypan blue exclusion dye (Thermo Fisher) was used to quantify the dead cells. The Spearman-Kärber method [33] was used to calculate the $TCID_{50}$ mL^{-1} for each well. The 95 % confidence limit, C ($C = \pm 2\ S_e$), was determined for each assay, where S_e is the standard deviation. In cases where the virus titer was undetectable by the $TCID_{50}$ assay, the large volume plating (LVP) assay was used to increase the sensitivity of the assay as given by Fan et al. [20].

2.7. MVM labeling and determination of virus retention profiles using LSCM

MVM particles were stained with DyLight™ 633 NHS Ester Fluorescent dye before spiking. Briefly, 1.2 mL of 11.10 logCopies mL^{-1} purified MVM particles (qPCR) were initially transferred to 100 kDa Amicon Ultra-15 Centrifugal Filter and buffer exchanged with 50–100 diavolumes of the labeling buffer (0.1 M sodium phosphate, 0.15 M NaCl at pH 7.36). The MVM particles were then mixed with fluorescent dye at a concentration of 1 g L^{-1}. The mixed MVM-dye solution was incubated for an hour under gentle agitation using a laboratory shaker. The labelled MVM particles were washed thoroughly with 100 diavolumes to remove any unbound dye molecules using the same labeling buffer in an Amicon Ultra-15 Centrifugal Filter. The feed reservoir containing labelled MVM was covered with aluminum foil to prevent photobleaching or degradation. All other conditions were the same as previously described.

After virus filtration, the hollow fibers were removed from the module and immediately incubated in a 4 % PFA solution overnight to immobilize the virus particles. The fibers were then washed with PBS solution, cut into small pieces, and embedded in Tissue-Tek O.C.T. Compound at −20 °C for sectioning. Fiber cross-sections with 8-μm thickness were cut using a Leica CM 1800 cryostat microtome (Leica Biosystems Nussloch GmbH, Buffalo Grove, IL, USA). Following cutting, the fibers were mounted on a glass slide, then covered with a coverslip and sealed using nail polish. The samples were imaged with a Leica TCS SP5 CLSM (Leica, Buffalo Grove, IL, USA) at 40X and 10X magnifications. The DyLight™ 633 NHS Ester dye was excited with a 638 nm (red) laser, with fluorescent emissions from 640 to 670 nm. Both brightfield (white light) and fluorescence images, as well as overlapping images, were collected using LAS X software (Leica) and analyzed with the ImageJ (ver. Fuji, Wayne Rasband, Washington, DC, USA) plot profile tool. Brightfield images were collected to define the positions of the fiber inner and outer surfaces. Signal intensities were normalized to a maximum of 100 and plotted against the relative membrane depth, with the inner surface assigned a value of 0 and the outer surface assigned a value of 1. The method described by Leisi et al. [26], which makes use of membrane autofluorescence, was to determine the variation of membrane porosity for the three virus filters. The excitation and emission wavelengths were 405 nm and 422 nm, respectively.

3. Results and discussion

3.1. Feed stream characterization

Table 2 gives the results of SEC analysis for the mAb and BSA solutions. The SEC chromatograms are given in Fig. S1 (supplementary material). As can be seen, the mAb shows little aggregation at the buffer condition tested here. However significant aggregation occurs for the BSA feed stream. Table 2 also gives the hydrodynamic diameter of mAb and BSA as measured by DLS. Fig. 3 gives the DLS intensity distribution. In the case of BSA, the average diameter is a little larger than expected (usually around 7 nm [34]). The measured average hydrodynamic diameter of the mAb is in the expected range.

3.2. Virus filtration

The variation of permeate flux with throughput (constant pressure filtration) as well as the variation of feed pressure with throughput (constant flux filtration) are given in Fig. 4(a–c). Fig. S2 (a-c) (supplementary material) give analogous data for non labelled MVM. The recommended operating conditions, particularly the maximum recommended operating pressure, differ for each of the three filters. Therefore, to compare the experimental results obtained from these different virus filters, the filtration conditions were standardized. Further as previous studies [20] indicate that low fluxes can lead to lower LRV, filters were run at the highest pressure for the filter with the lowest pressure rating. Results are given for filtration runs with buffer, 5 g L^{-1} mAb and 5 gL-1 BSA spiked with labelled MVM. Results for constant pressure filtration are read using the left-hand side y-axis and are given by the solid curves while results for constant flux filtration are read using the right-hand side y-axis and are given by the dashed curves.

LRVs for constant pressure filtration for feed solutions spiked with non labelled and labelled MVM are given in Fig. 5. Analogous data for constant flux filtration are given in Fig. 6. Tables S1–2 (supplementary material) provide the same data in tabular form. Virus feed titer was determined by the $TCID_{50}$ assay. Both figures indicate that under all conditions for all three feed streams the LRVs for all three virus filters were more than 4 logs. Further there is no difference in LRV between the feed solutions spiked with labelled or non labelled MVM. This is not surprising as the dye has a molecular weight (MW) of 1066 Da according to the manufacturer, insignificant compared to the MW of MVM at ~4 MDa. Further as the dye is hydrophilic it is unlikely to lead to virus aggregation. Consequently, the LRV for non labelled MVM at constant flux were not determined for the 20 N filter. As no virus particles were detected in the permeate and buffer chase fractions with $TCID_{50}$ assay, LVP assay was performed to improve the sensitivity of the measurement. As can be seen virus breakthrough was only detected for the 20 N filter for all feed streams and the S20 N filter for BSA feed solutions under constant pressure filtration for both labelled and non labelled virus. LVP results indicated that there was virus breakthrough in each fraction and the buffer chase for the 20 N filter. For the S20 N filter, virus particles were detected only in fraction 3.

Fig. 7 and Fig. 8 give the mAb and BSA recovery for constant pressure and constant flux filtration respectively for feed solutions spiked with labelled and non labelled MVM. Tables S3–4 (supplementary material) provide the same data in tabular form. In all cases the recovery was close to or equal to 100 %. The presence of labelled MVM had no effect on recovery. Given the similarity in the results for constant pressure and flux filtration, mAb and BSA recovery were not determined for non labelled MVM for the 20 N filter for constant flux filtration.

Table 2
Analysis of feed streams.

Feed stream	Monomer (%)	Dimer (%)	Trimer (%)	Hydrodynamic diameter (nm)
BSA (5 g L^{-1}, PBS buffer)	84.9	13.2	1.9	9.7 ± 0.028
mAb (5 g L^{-1}, PBS buffer)	99.1	0.87	0.0	11.4 ± 0.012

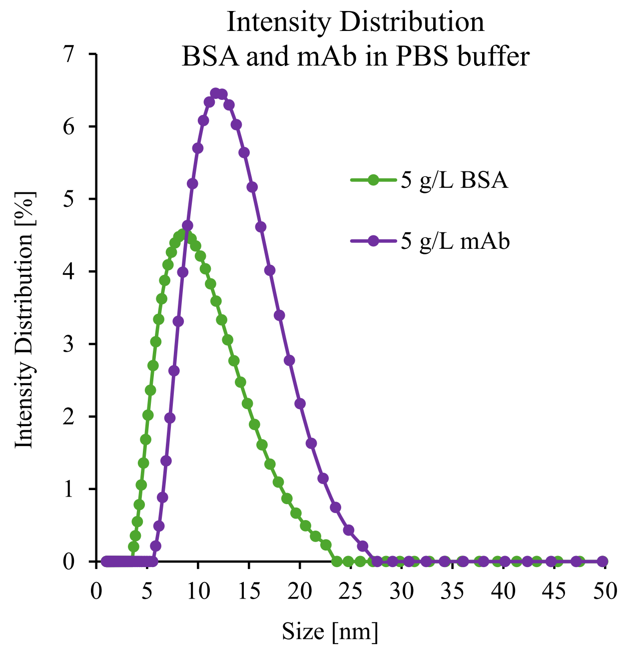

Fig. 3. Hydrodynamic diameter of mAb and BSA determined by DLS.

3.3. LSCM imaging

The location of virus entrapment as determined by LSCM is given in Fig. 9(a–c) where (a), (b) and (c) give results for feed streams containing MVM, MVM and mAb and MVM and BSA for the 20 N, S20 N and BioEX virus filters. Results for constant pressure are given by solid curves while results for constant flux are given by dashed curves. The results indicate that the location of virus entrapment depends on the membrane, feed properties and filtration conditions.

4. Discussion

The hydrodynamic diameters of BSA and mAb measured by DLS show a distribution of sizes. In the case of BSA the distribution of diameters as well as the large than expected average hydrodynamic diameter is most likely due to the presence of irreversible aggregates. The presence of irreversible aggregates is confirmed by the SEC results. For the mAb the amount of irreversible aggregates (less than 1 %) is very low. Further the hydrodynamic diameter agrees with the hydrodynamic diameter for mAbs [11]. However, the range in hydrodynamic diameter based on the intensity distribution (Fig. 3) indicates there are a significant number of mAb species with hydrodynamic diameters larger than the average value.

It is known that protein-protein interaction can be strongly affected by the solution condition. In the case of BSA with an isoelectric point (pI) of 4.7, it will be negatively charged at pH 7.36. Negatively charged BSA will have repulsive electrostatic interaction. However, this electrostatic repulsion will be screened in the presence of salt ions in the solution, particularly at salt concentration over 0.3 M NaCl based on previous studies [35,36]. Higher salt concentration and highly charged ions lead to slightly attractive interaction between the BSA molecules. Here the buffer solution contains 0.137 M NaCl and 0.0081 M sodium phosphate which could lead to attractive protein-protein interactions and thereby an increased hydrodynamics diameter.

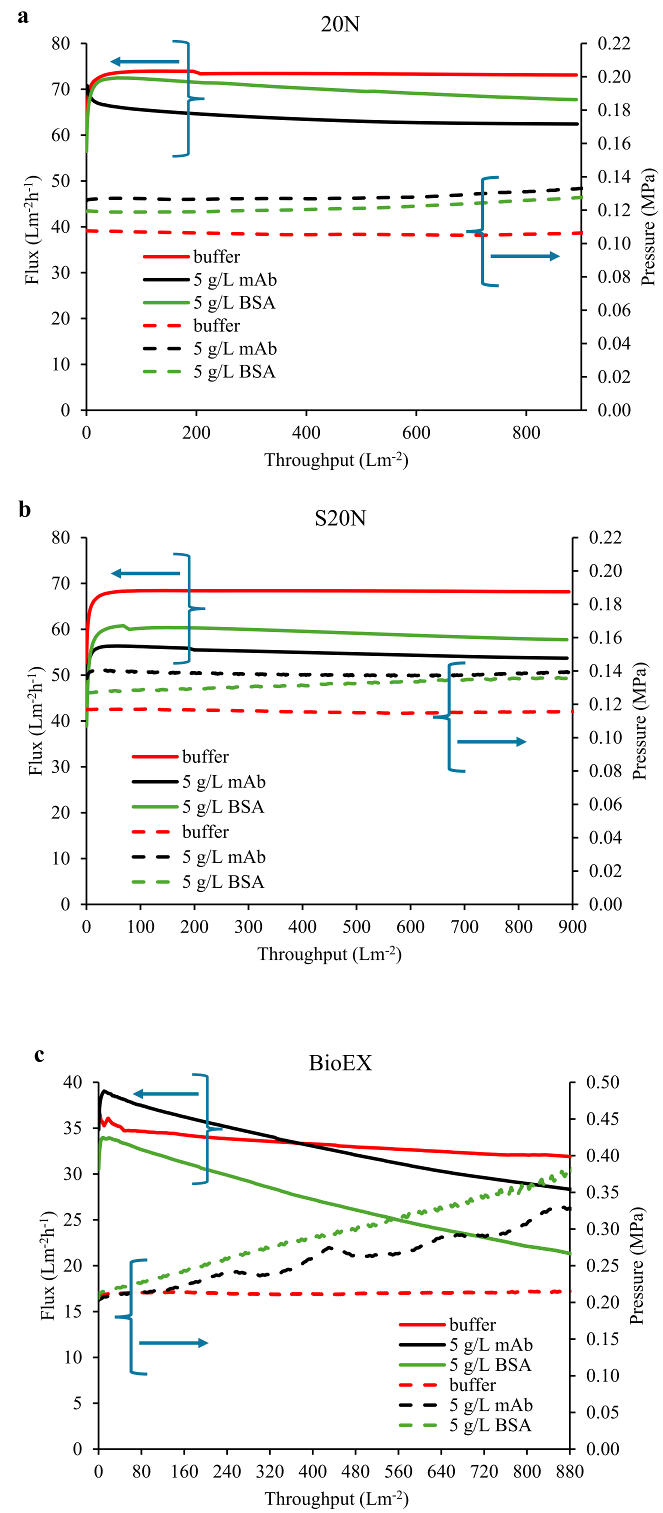

Fig. 4. (a–c) Flux and pressure profiles for filtrations spiked with labelled MVM. Fig. 4(a), (b) and (c) give results for 20 N, S20 N and BioEX respectively for feed streams consisting of buffer, 5 g L^{-1} mAb, and 5 g L^{-1} BSA spiked with MVM. Results for constant pressure filtration are read using the left-hand side y-axis and are represented by solid curves. Results for constant flux filtration are read using the right-hand side y-axis and are given by dashed curves. Note that the 20 N and constant flux run for BioEX are operated at pressures slightly above the manufacturer's specification.

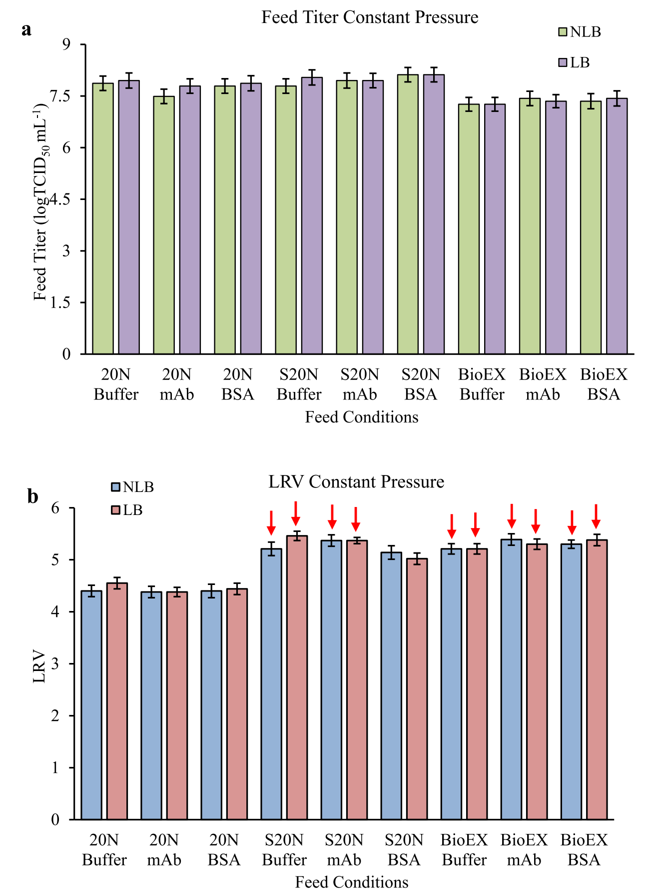

Fig. 5. a) Feed titer and b) LRV for labelled (LB) and non labelled (NLB) MVM for constant pressure filtration. Red arrow indicates that no virus was detected in any fractions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Similarly, the self-assembly of the native mAb can lead to the formation of reversible or soluble oligomers. Unlike irreversible aggregates, they are held together by weak intermolecular interactions. They cannot be detected by SEC. However, they lead to a distribution of hydrodynamic diameters as determined by DLS. Evidence of the presence of reversible mAb aggregates is given by the fact that a distribution of hydrodynamic dimeters is observed by DLS yet undetected by SEC [37].

Virus filtration was conducted under constant pressure and constant flux conditions. The manufacturer's recommended maximum feed pressures for the 20 N, S20 N and the BioEX filters are: 0.098 MPa (14.2 psi); 0.216 MPa (31.3 psi) and 0.343 MPa (49.7 psi). In order to test all three filters under the same conditions the feed pressure was set at 0.1 MPa. For constant flux operation, a flux of 75 L m^{-2} h^{-1} was chosen for all three filters. This resulted in the flux for constant pressure operation being only a little less than the flux for constant flux operation for the 20 N filter with MVM spiked in buffer. However, for mAb and BSA filtration with S20 N and BioEX filters, the differences in flux for two modes of operations are greater.

The flux and pressure profiles given in Fig. 4(a) and (b) are similar for the 20 N and S20 N filters. This is not surprising as the membranes are similar, both being regenerated cellulose. The S20 N is a newer version of the 20 N filter. S20 N is slightly thicker with a higher mechanical strength than 20 N [38]. The figures indicate that there is little fouling of the filters as both the pressure and flux profiles are almost flat. The S20 N is a higher-pressure filter. Further, the resistance of the membrane to flow is higher hence the fluxes under 0.1 MPa are a little lower and the pressures at constant flux are slightly higher than for the

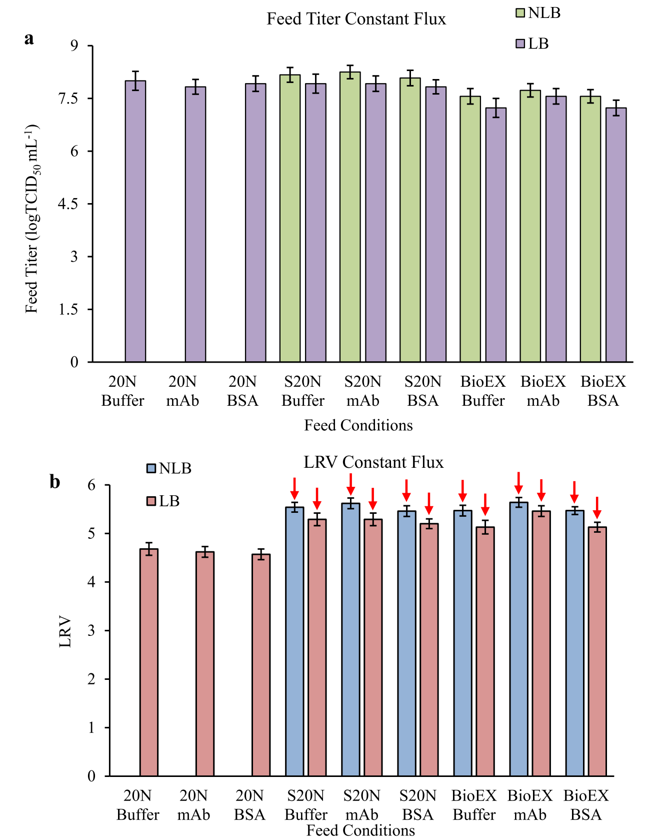

Fig. 6. a) Feed titer and b) LRV for labelled (LB) and non labelled (NLB) MVM for constant flux filtrations. Red arrow indicates that no virus was detected in any fractions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

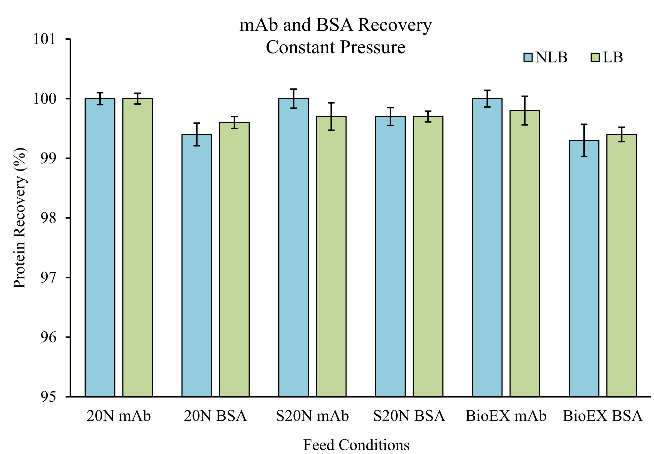

Fig. 7. mAb and BSA recovery for constant pressure filtration spiked with labelled or non labelled MVM.

20 N.

The results for both filters indicate that buffer spiked with MVM (red curves) has the least fouling with highest flux at constant pressure and lowest pressure at constant flux. Moreover, the flux and pressure profiles are more or less constant during the course of the 900 L m^{-2} filtration. These results provide evidence of the purity of the virus stock. The flux and pressure profiles given in Fig. 4 are for labelled MVM. Analogous flux and pressure profiles for non labelled MVM are given in Figs. S2 (a-

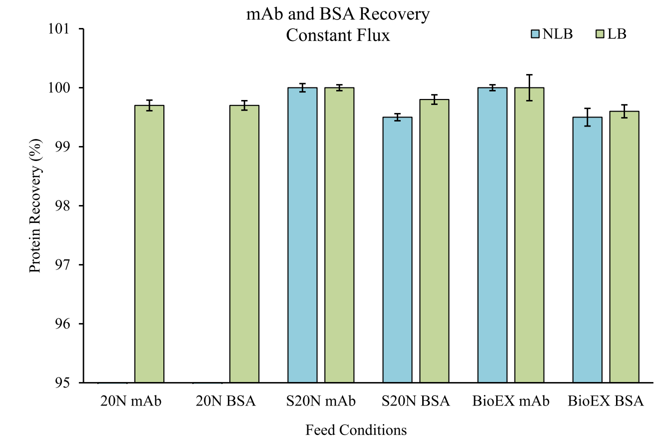

Fig. 8. mAb and BSA recovery for constant flux filtration spiked with labelled or non labelled MVM.

c) (supplementary material) and indicate labelling has no observable effect. Fig. 4 (a) and (b) indicate that the flux (at constant pressure) decreases for feed streams in order of buffer, BSA and mAb for the 20 N and S20 N filters. The results for constant flux are analogous. The pressure increases for feed streams in order of buffer, BSA and mAb. Fouling for all feed streams is insignificant.

The flux and pressure profiles indicate significant fouling for the BioEX filter as shown in Fig. 4 (c). Unlike the 20 N and S20 N filters, this filter is made of hydrophilically-modified polyvinylidene fluoride. Consequently, the membrane surface properties and morphology will be different. The result highlights the important influence of membrane properties on performance. Nevertheless, analogous to the 20 N and S20 N filters, the feed stream containing buffer shows the least fouling. In the case of constant pressure filtration, it appears the flux for buffer is a little lower initially than the mAb feed stream, but the two curves cross over at a throughput of about 380 L m^{-2}. This difference is probably due to slight experimental variability during start up.

Unlike the 20 N and S20 N the BSA feed stream for the BioEX filter shows the greatest fouling. The results highlight that the effect of protein-membrane interaction is highly material specific. Figs. 5 and Figs. 6 indicate that for all three feed streams for the operating conditions considered here, the LRVs are more than 4 for all filters. Being a newer version of the 20 N, the S20 N provides more robust virus clearance.

Virus filter performance will depend on the membrane structure especially the membrane surface properties. Fig. 9 (a-c) provide unique insights into the location of virus entrapment for the three virus filters. The original confocal images are given in Fig. S3–5 (supplementary information). The results are significant as they indicate changes in virus entrapment under operating conditions that are of industrial relevance providing a basis for comparison of the three filters. The results presented here are for experiments that were run for long times. For constant flux operation, run times were 12 h. For constant pressure operation, run times were 12.5–14 h for 20 N, 13.5–16 h for S20 N and 26–40 h for BioEX. The following general observations may be made for all three filters.

Constant pressure operation tends to lead to virus entrapment deeper with the membrane compared to constant flux operation due to the reduced flux and increased filtration time. Fig. 4a–c indicate that the fluxes are higher for constant flux operation compared to constant pressure operation. Our previous results [20] show that low flux around 5 L m^{-2} h^{-1}, when diffusion is more dominant, leads to significant virus breakthrough for 3 out of 4 commercially available virus filters. However, at an increased flux of 40 L m^{-2} h^{-1} when convective flow becomes prominent, for the identical feed solution, little or no virus breakthrough

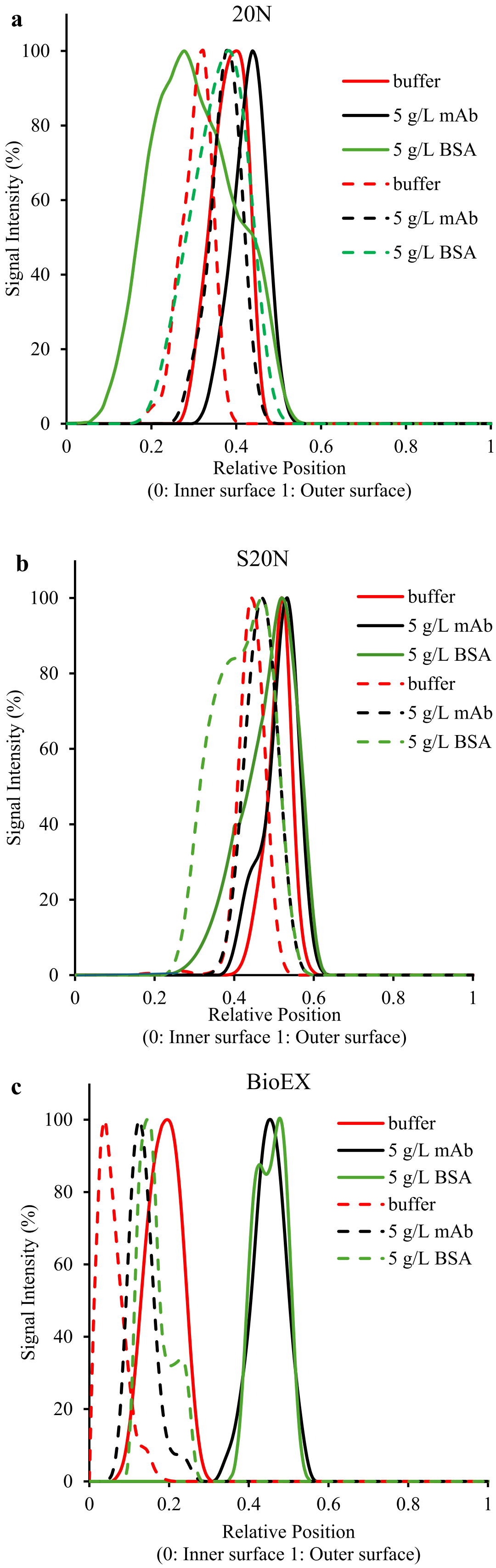

Fig. 9. (a–c) Virus capture profiles for (a) 20 N(b) S20 N and (c) BioEX, respectively.

was observed. Afzal and Zydney [23] also investigated the effect of permeate flux on virus titer in the permeate for a much more homogeneous membrane. They found that lower permeate fluxes lead to a decrease in LRV. Their model virus was a bacteriophage, and they measured virus titer in the permeate. For the experiments conducted here, though there was little virus breakthrough in the permeate, the trend is the same, greater virus penetration at lower fluxes for all three membranes. In addition, the data obtained here were for long run times where diffusion can play a more important role.

The location of virus entrapment in the membrane will depend upon membrane properties such as asymmetry and porosity. The manufacturer does not release detailed cross-sectional images illustrating the structure of Planova filters. Fig. S6 (supplementary material) is a scanning electron micrograph (SEM) image of a 20 N hollow fiber provided by the manufacturer. Though no scale is given it clearly indicates that not only is the membrane asymmetric there is also a variation of porosity with position in the membrane [39]. The 20 N and S20 N contain regenerated cellulose hollow fibers. Both will display similar asymmetry and variation of porosity with position in the membrane. Suh et al. [22] provide SEM images that clearly indicate the asymmetric structure of the 20 N and BioEX membranes. It should be noted that all Planova filters are manufactured via wet spinning using biaxial stretching. Since the poor solvent permeates from the surface, this results in an asymmetric structure like that of the Planova 20 N filter.

We have used the method described by Leisi et al. [26], which makes use of membrane autofluorescence to determine the variation of membrane porosity for the three virus filters. The data were smoothed using a 5th order polynomial. Fig. 10 gives the variation of the relative autofluorescence with position for the three membranes. A higher level of autofluorescence indicates lower porosity. It is important to note that one cannot quantitatively compare the relative intensity curves for different membranes. However, the shapes of the curves highlight differences between the variation of porosity with position for the different membranes. Relating Fig. 10 to Fig. 9(c), it can be seen that the BioEX membrane has a lower initial porosity (higher relative intensity) compared to the porosity variation for the other membranes explaining the fact that the capture zone is closer to the feed side of the membrane.

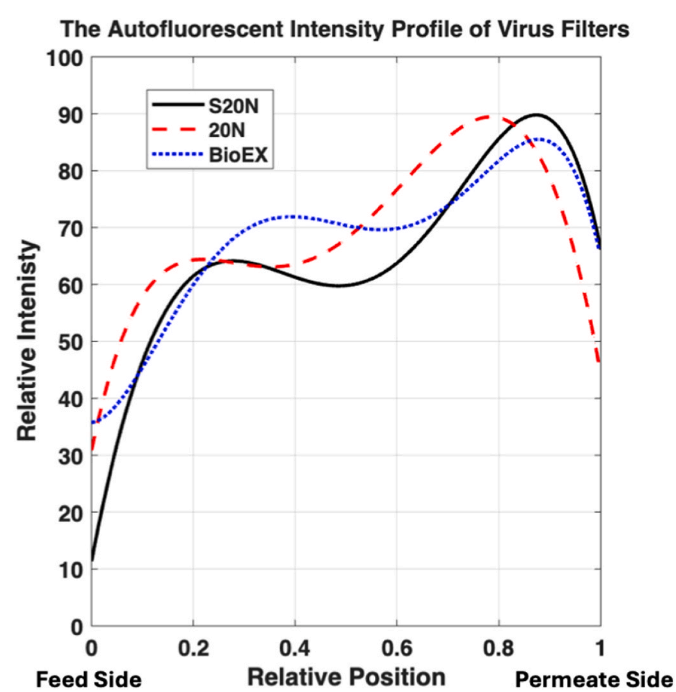

Fig. 10. Variation of relative autofluorescence intensity with position in the membrane.

The variation of the relative intensity for the 20 N and S20 N is similar though the S20 N is a thicker membrane. Fig. 10 provides a qualitative basis for the changes in virus capture zones observed in Fig. 9.

The presence of a mAb leads to entrapment of virus particles deeper in the membrane compared to the buffer only condition as shown in Fig. 9(a–c). Moreover, no broadening of the entrapment zone is observed. By directly measuring the virus titer in the permeate, Afzal and Zydney [23] demonstrate that proteins can displace previously entrapped viruses in a virus filtration membrane. Several factors can contribute to this effect. Firstly, mAb molecules interact and adsorb onto the membrane pore surfaces leading to the narrowing of the membrane pores, which in turn could force the virus particles to migrate into larger pores leading to deeper entrapment. The penetration of virus particles deeper into the filter in the presence of mAb is more pronounced at lower flux when filtration was run at constant pressure for all three filters. As a result of lower flux and longer filtration times diffusion becomes more important. Secondly, an enhanced reversible aggregation of the mAb molecules inside the membrane pore could also lead to virus displacement. hIgG used in earlier work by Afzal and Zydney [23] is known to form reversible aggregates. Esfandiary et al. [40] indicate that even at low concentrations, mAbs can form weak reversible aggregates. Most reported studies have considered reversible aggregation under quiescent conditions. However, within the pores of a virus filter, mAb molecules will be brought very close to each other under an applied shear stress. This could lead to enhanced self-aggregation together with localized changes in fluid properties. Reversible aggregates can also displace virus particles deeper into the membrane. Here we show for first time that the presence of mAb with little irreversible aggregation leads to the migration of MVM particles deeper into the membrane towards the permeate side.

Fig. 9 (a-c) indicate that while the presence of both the mAb and BSA leads to migration of the virus capture zone deeper into the membrane, BSA leads to significant broadening of this zone. In the case of BSA, SEC analysis indicates the presence of over 10 % irreversible aggregates. Past studies [22,23] indicate that the presence of irreversible aggregates can lead to decreased as well as enhanced LRV. The effects depend on the distribution of aggregate sizes relative to the virus particles and the membrane structure. Here we show for the first time that in the case of BSA, the presence of aggregates similar in size to the virus particles can lead to entrapment of virus particle deeper in the membrane structure. This is probably due to the fact that aggregate will be trapped in location where virus would have been trapped allowing viruses to pass through larger pores further into the membrane. Further the entrapment zone broadens probably due to the interactions between the aggregates and the virus particles. The larger protein-virus aggregates can be rejected closer to the feed side of the filter leading to the observed broadening of the entrapment zone.

The results for the 20 N and S20 N are insightful as there is little variation in permeate flux or feed pressure for both constant flux and pressure operations. Consequently, the membrane resistance is constant. The results for the BioEX filter provide insights into how changes in membrane structure can affect virus entrapment. The unique BioEX membrane structure results in virus entrapment very close to the feed side of the membrane. However, the migration of the entrapment zone into the membrane in the presence of mAb and BSA is much greater than for the 20 N and S20 N, particularly under constant pressure filtration conditions, where significant flux decay was observed leading to a much increased filtration time. As mentioned previously, longer run times favor diffusive migration of virus particles which leads to their entrapment deeper in the filter. Nevertheless, it should be noted that even though migration of the entrapment zone is greatest for the BioEX filter, it still maintains the best virus clearance for all the filters tested.

Taken together the result obtained here provide further insight into the location of virus entrapment during virus filtration under industrially relevant conditions. Clear evidence is provided on the effect of a product species on the location of virus entrapment due to its interaction with the membrane surface and the presence of irreversible and reversible aggregates. Further evidence is provided that the filtration time taken to process the feed and permeate flux can affect the location of virus entrapment. These insights will be important when designing virus filters for low flux high product concentration operation as well as for long run times in continuous biomanufacturing operations.

5. Conclusion

For the first time detailed visualization, using labelled MVM and LSCM, of virus entrapment in virus filters during operation has been conducted over a range of industrially relevant operating conditions. Three commercially available hollow fiber membranes consisting of regenerated cellulose and polyvinylidene fluoride were investigated. Both constant pressure and constant flux operations were investigated. The filters were run under industrially relevant conditions that did not involve overloading the virus filters to induce virus breakthrough in the permeate. Control experiments indicated that labelling MVM had little influence on filter performance. The results indicate that irrespective of whether the virus filters are run under constant pressure or constant flux conditions, lower permeate fluxes lead to entrapment of the virus particle deeper within the membrane.

The entrapment profiles for the two regenerated cellulose membranes, 20 N and S20 N are similar while those of the polyvinylidene fluoride membrane, BioEX, are different highlighting the importance of membrane material and properties on the location of virus entrapment. The throughput for all experiments was about 900 L m^{-2}. For the S20 and S20 N, little variation of flux or pressure was observed. Consequently, the resistance to permeate flow was constant during operation. However, for the BioEX filter, the resistance to permeate flow increased during operation. Changes in resistance to permeate flow during operation will also impact the location of virus entrapment. The results obtained here suggest that fouling can lead to further displacement of virus particles deeper into the membrane.

The presence of a product species in the feed stream has a significant effect on the location of virus entrapment. The presence of a mAb that does not display significant irreversible aggregation leads to displacement of the virus particles deeper into the membrane due to their interactions with membrane pore surface and the formation of reversible aggregates. The presence of BSA that forms irreversible aggregates leads not only to displacement of virus particles deeper into the membrane but also leads to a broadening of the entrapment zone. These observations apply to all three virus filters. However, in all cases the virus filters displayed at least 4 LRV. The results are representative of the location of virus entrapment during operation.

As indicated in our previous work [20] operating virus filters at very low fluxes where the Peclet number is $\ll 1$ (diffusion dominates), leads to a significant reduction of LRV. Consequently, hydrodynamic conditions can also play a critical role in performance. For the experiments performed here, the throughput was kept the same at 900 L m^{-2} for all the filters> A lower flux means a longer filtration time which varied from 12 h to ~40 h. For 20 N and S20 N, the difference in flux is small between the constant flux and constant pressure modes. The virus capture zone moved only slightly to the permeate side for longer filtration (lower flux) conditions. The flux difference becomes more marked for BioEX between the two modes, where the largest migration of the capture zone was observed at lower flux conditions. Here we aimed to minimize any hydrodynamic effects.

Taken together, the results obtained here provide new insights into the performance of virus filters. There is a great need for cost reduction in the manufacture of biopharmaceuticals. The cost of the downstream purification operations can be 70 % or more of the total manufacturing cost. Development of intensified purification operations will be essential. However, development of intensified purification trains could result in the need to purify high concentration mAb streams. Further continuous biomanufacturing processes will result in lower flow rates but much longer run times for virus filters [41,42]. This results in challenging operating conditions for virus filters.

In continuous biomanufacturing operations, it is essential the product stream from the unit operation just upstream from the virus filter (frequently a membrane adsorber run in flow through mode as a polishing step) be introduced directly to the virus filter. Flow rates to the virus filter will be lower and product concentrations higher than is the case in current batch processing [29]. The results obtained here suggest that under these conditions virus particles can migrate deeper into the membrane before capture. Movement of the virus capture zone further into the membrane structure could lead to lower LRV as virus particles may pass through the membrane. Membranes like the BioEx where the initial porosity is lower will be more robust under these conditions. A lower porosity will require higher operating pressures for a given flux. However, for continuous biomanufacturing processes, fluxes will be lower. The required permeate flux will depend on the flow rate from the previous unit operation, most likely the flow through from a membrane adsorber.

CRediT authorship contribution statement

Wenbo Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xianghong Qian:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Hironobu Shiratak:** Writing – review & editing, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Daniel Strauss:** Writing – review & editing, Resources. **Ranil Wickramasinghe:** Writing – review & editing, Writing – original draft, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ranil Wickramasinghe reports financial support was provided by Asahi Kasei Life Science Co. Xianghong Qian reports financial support was provided by Asahi Kasei Life Science Co. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Funding for this work was provided by Asahi Kasei Life Science Co, Tokyo, Japan.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124781.

Data availability

Data will be made available on request.

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